

Detecting Memorization in LLM Generated Music

1st Jacob Edward Galajda

College of Engineering and Computer Science
University of Central Florida
Oviedo, United States
jacob.galajda@ucf.edu

2nd Kien A. Hua

College of Engineering and Computer Science
University of Central Florida
Oviedo, United States
kien.hua@ucf.edu

Abstract—Music plagiarism detection is a challenge that pre-dates artificial intelligence. As music generation becomes more accurate and popular, systems to detect biases and memorization are necessary to ensure fair use and promote transparency. In this paper, we introduce a set of standards and protocols to detect memorization and stylistic bias in an unsupervised data setting. Applying this model agnostic approach to a subset of LLM generated music identified that self-similarity within LLMs exceeded that of the compared human baseline by a factor of over 200. Further, this approach detected memorization within LLM generated output, however, more information regarding pre-training datasets is required to identify which local minimum the LLM may be gravitating towards.

Index Terms—symbolic music, language model, bias detection

I. INTRODUCTION

Within the past couple of years, Large Language Models (LLMs) have adapted rapidly and expanded their base knowledge; leaving many users and researchers alike to question the origin of the data and how original the responses become. Substantial work has gone into music recognition [1]–[3] for applications like song recommendation, personalization and automated genre classification. Some attempts aim to solve composer classification [4], visual analysis [5], [6] and recent attempts in music plagiarism [7] to detect when humans plagiarize one another in music. New techniques [8] already compare audio waveform outputs from platforms like Suno to human samples. With recent litigation surrounding LLMs and human composed songs, music generation research needs the capability to automatically detect similarity within training data; and also improve music generation quality.

Fields such as Natural Language Processing (NLP) have substantial progress in not only plagiarism detection but also thematic summarization. Individual words can have various meanings and across different contexts, necessitating for deep learning solutions to accurately classify and interpret the nuanced meaning of each phrase [9]. When applied across sequences, author detection and plagiarism applications [10] become more achievable as the written medium has derived signals dedicated for this task. With these techniques applied to create *music as a string* [11], we’re able to yield a human to human baseline comparison with NLP. With promising new evidence found in training data extraction [12], [13] and document reasoning [14], [15], there is a compelling argument that LLMs are in fact generating residual data from training.

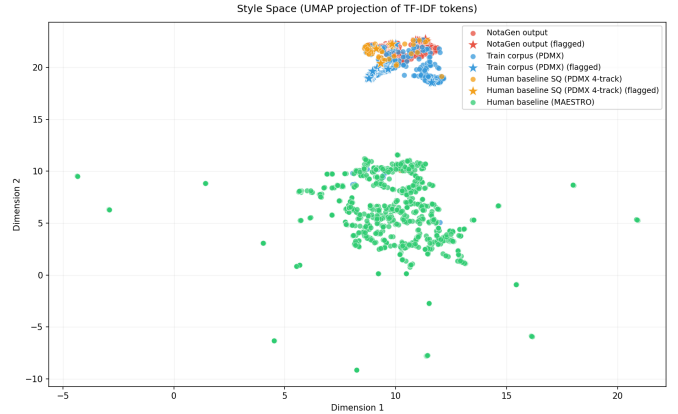


Fig. 1. UMAP projection of the TF IDF token embeddings. Stars indicate pieces that exceed calibrated similarity thresholds. The clean separation of green dots (piano roll) confirm instrumentation variations and validates the dual-baseline experiment design.

But with only binary classification and no existing statistical significance testing, these techniques are not yet suited for determining originality across LLM generated outputs and comparing training influences with human samples.

The underlying research for this paper was motivated by a previous experiment [16] in which some model configurations generated eerily familiar to songs outside of the published dataset. Previously, this LLM generated batches of string quartet symbolic music with a subset of fine tuning data from the PDMX dataset [17]. During the human survey, several participants noted listening to samples that sounded seemingly identical to works they have heard before from human composers. While embedding algorithms like CLaMP [18] are commonly used to distinguish works and promote diversity in generated samples, the discrepancies noted in the prior human survey group prompted this research to identify techniques and strategies to discern this feedback. Thus, we present the following contributions to further the dialogue between musicians and AI music researchers:

- An unsupervised utility capable of matching interpretable evidence of style transfer and bias detection across generated samples and suspected corpora
- Double null-baseline design to showcase discrepancies across instrumentation; highlighting the significance in

not only tokenization detection but also medium in generated music

- Pareto front threshold calibration to empirically determine thresholds from the null distribution

II. RELATED WORKS

Detecting plagiarism within music is non-trivial to automate. However, it is a necessary step for detecting memorization as the traces modified or repeated by the music generation system must have a human sample to be identified by. However, previous works have focused largely on individual human to human detection of plagiarism. Expanding on previous works in plagiarism detection is essential for creating a system capable of accurately detecting memorization at aggregate and individual scales.

A. Memorization within Large Language Models

Many memorization studies have been conducted in the Natural Language Processing space. Notable contributions from Carlini et al. [12], [13] showcase various prompt attacks to recover memorized training data without having direct access to the model itself. While exact to semantically similar content varies, this research showcases that LLMs can be subjected to repeat sequences found within their training. In context to music generation research, the implications of this could entail repeating a score used during various phases such as pretraining or fine tuning.

In order to detect memorization within music, techniques must be established to analyze verbatim and non-discrete methodologies of memorization; as music has many nuances that are apparent to human listeners but non-trivial for algorithms to detect. Additionally, the distinction between audio and symbolic music must also be made as the available signals within each require different data ingestion strategies altogether. However, common NLP techniques can accurately apply to memorization detection tasks with careful regard to data source normalization.

B. Similarity Metrics for Music Generation

There are many parallels between music plagiarism, music memorization detection, composer analysis and music recommendation systems. Detecting similarity heuristically [4], [19] can be accomplished for composer classification, however, this task was historically framed to classify whole songs rather than individual nuances. When applied to music summarization and recommendation systems [20]–[22], much emphasis is applied to developing core units for similarity comparison. Visualizations of symbolic sheet music [5] and audio waveform files [23] are also considered for comparing the music as each piece develops through time. To properly conduct memorization research in music, facets from each of these fields must be considered.

Structural metrics like rhythmic patterns and pitch detection are essential for identifying the structural components of music. With concepts heavily rooted in NLP like n-gram phrase overlap, analyzing symbolic music has become a very

predictable practice in music recognition. Recently, De Prisco et al. [7] introduce an interval-based melody representations using vector embeddings with semitone intervals and duration features. With this, systems can detect transposed pieces of music which may cause two songs to sound different while both still share the same structure.

Malandrino et al. [11] further this result by applying NLP techniques to create *MaaS*, or music-as-a-string, which enables the use of n-gram metrics, yielding high detection accuracies. One novelty in this approach is the orchestration of metrics across a lighter representation of music. However, asymmetric length is not considered and detecting plagiarism from two humans does not require calibration as does memorization detection across a wide pretraining dataset.

With a labeled and notated dataset, [24] utilize audio waveform features to extend the structural analysis through motion. Similar to previous approaches, segments are computed and a pairwise evaluation is considered. While the Top-1 accuracy is substantially lifted when considering their multi-metric approach, the quantity of data and limitations to the labeled data prove difficult for yielding applicable results for memorization detection. In MelodySim [25], MIDI files are synthesized into waveform and achieve a remarkable 97% accuracy within dataset. While directly suited for analyzing songs from streaming platforms, the extensive training time and dataset pruning make it difficult to rapidly integrate and scale this technique.

With these considerations, some researchers look to relationship data and visualization for similarity detection. Hernandez-Olivan & Beltran [26] utilize a graph-based approach to conduct low-cost retrievals across a wide variety of music. Galajda et al. [5] propose visualizing composer discographies side by side, however, the visualizations are difficult to interpret and individual comparisons are not easily extracted from such comparisons. Across these strategies, the embedding approaches like CLaMP [18] have emerged as extraordinarily popular tools for downstream tasks like LLM reinforcement learning and genre detection. One commonality all of these approaches share is that each provide a substantial global basis for broad thematic elements and reduced support for accurate, centralized detection of finer details necessary for memorization detection. In the case of embedding representations, nuanced changes may disproportionately effect similarity like in hash functions. Identifying which elements of the music in question match that of the corpus are not easily answered with the previous approaches. Thus, this research focuses on unsupervised memorization detection that supports large quantities of songs with low computing requirements.

III. METHODOLOGY

In order to quantify plagiarism in music compositions, we apply a model agnostic approach to a single, known instance of copying to assess the viability of this theory. More music generation models should be studied in future works with this framework.

TABLE I
CORPUS GROUPS

Group	n	Role
LLM output	219	Test group (NotaGen SQ)
Fine-tune proxy	50	Instrumentation-matched null
Train corpus	300	Broad PDMX proxy
MAESTRO	1,276	Cross-domain null

A. Dataset

To test the memorization, LLM outputs and suspected human samples are required. The LLM output yielded from [16] experiment are used as a baseline for the LLM. Two human-composed baseline datasets are used for the experiment. PDMX [17], a well-known MusicXML formatted dataset with available MIDI files, contains various string quartet and quartet samples. While it is not specific which datasets were used in the pre-training process of Notagen, the PDMX entries were used at some point in the fine tuning process of the previous experiment; making this experiment highly relevant.

The Maestro dataset [27] is included as a necessary secondary baseline to showcase style discrepancies within the metrics. One limitation to this research is the instrumentation used across songs. Analyzing the string quartet rendition of a pop song will produce different results and should be studied in future works. Finally, a fine-tune proxy subset is selected from PDMX to ensure samples were quartet arrangements. A full distribution can be found in Table I.

B. Tokenization

To preserve data integrity, input data is restricted to the MIDI file format and interpreted using the Music21 Python module [28]. Pitch duration tokens follow the algorithm and pitch-interval with normalized length (PINL) representation found in [11]. Briefly, the algorithm computes the minimum duration found within the corpus for dynamic normalization. N-grams from these tokens later are constructed for pairwise evaluation. Semitone intervals are also computed as in [29] to cover a larger surface area of detection.

C. Metrics

We hypothesize memorization has multiple signals and using a single metric does not accurately depict memorization. Previous studies [1] indicate a variety of metrics yielded the best results to identify more nuances found within music. Music structure is a very obvious sign of similarity, however, similarity and memorization can be exhibited outside of music structure.

1) *Jaccard Overlap*: N-gram Jaccard overlap is very effective for complex multimedia comparison tasks [30] and is utilized for immediate matching of consecutive tokens:

$$J_n(Q, R) = \frac{|G_n(Q) \cap G_n(R)|}{|G_n(Q) \cup G_n(R)|}$$

TABLE II
AVERAGE STYLE COLLAPSE (SC) BETWEEN GROUPS, SC_T = TF-IDF COSINE DISTANCE; SC_H = PITCH-CLASS HISTOGRAM COSINE DISTANCE. LCS = BEST-WINDOW LCS.

Group	n	Len	SC_T	SC_H	LCS
LLM	219	1102	0.702	0.258	0.0095
Fine-tune	50	617	0.721	0.283	0.0032
PDMX	300	472	0.842	0.398	0.0027
MAESTRO	1276	3143	0.518	0.194	0.0003

Where $G_n()$ denotes the set of distinct n-grams, query sequences denoted as Q and R reference sequences. For Jaccard overlap J_n with $n \in \{3, 5, 8\}$, we make note of overlap when all consecutive n tokens of the query match that of the reference. This method is highly effective at identifying verbatim copying. From separate testing, $n = 5$ showed the lowest false positive and highest true positive results.

2) *Best-Window LCS*: Longest Common Subsequence (LCS) is a well known approach to directly comparing similar array elements against two sources. Rather than scoring similarity as a discrete integer found in standard LCS tasks, the best scoring sequence lengths are considered; as the model can produce portions of training data at unknown lengths. Using the same notation standards as above, the following depicts the best window LCS:

$$LCS_{bw}(Q, R) = \max_k \frac{LCS(Q, R[k : k + |Q|])}{|Q|}$$

The normalization applied by $|Q|$ mitigates variable length songs found throughout music datasets. This technique is applied independently to evaluate absolute pitch for verbatim detection and intervals to cover melodic contour. This allows for transposition detection and identifying more subtle nuances in music that the Jaccard metric may not be able to detect.

3) *TF-IDF Cosine Distance*: Term Frequency-Inverse Document Frequency (TF-IDF) is another element that is useful for screening common trends while eliminating frequency biases in detection. In music, common phrases or expressions may be repeated across works and should not be emphasized for stylistic comparisons. However, unique portions of a song should be considered. For the purposes of this evaluation, each constructed n-gram is projected to 5,000 features to more accurately encapsulate the symbolic music without losing meaning. This fitting is also done to reflect the stylistic differences between the piano baseline and the string quartet samples.

One of the weak points in LCS is the temporal ordering of data. With TF-IDF in conjunction to LCS, additional coverage can be introduced to provide resilience to sequences occurring at different portions of a song. Style collapse can also be plotted and observed, such as in Figure 2. Using TF-IDF, we are able to detect frequently reused sequences and measure the overall repetition. When projected unto a two-dimensional surface using UMAP [31], visual similarities in style and

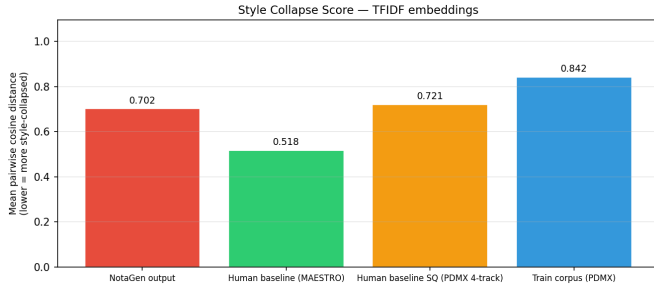


Fig. 2. The mean intra-group pairwise TF-IDF cosine distance (style collapse scores) for each representative corpus group. Note that higher cosine distances indicate greater diversity within the respective group.

TABLE III
PARETO-CALIBRATED THRESHOLDS

Metric	Dir	Threshold
<i>Abs-pitch</i>		
Jaccard $n=5$	>	0.0220
LCS strict	>	0.3874
TF-IDF dist	<	0.1332
Histogram dist	<	0.0010
<i>Interval</i>		
LCS interval	>	0.3874

denote the differences in style from the two included baselines. Figure 1 illustrates this as well as songs that were flagged for similarity above the calibrated thresholds.

D. Metric Threshold Calibration

Utilizing Pareto threshold calibration from [11], each metric computed yields a null distribution score from the provided human baseline pairs and the LLM to training corpus pairs. One in a thousand sampling is conducted across the evenly distributed samples to reduce computational overhead while providing enough of a baseline for threshold calibration. For each candidate threshold T , a false positive rate and true positive rate are calculated to the Pareto-optimal value; such that the false positive rate is reduced. Accordingly, Table III contains the computed thresholds for this experiment. LCS set to 0.387, Jaccard $n = 5$ to 0.0220 and TF-IDF to 0.1332.

To ensure validity of the Pareto calibration, five randomized runs were conducted on the dataset. These calibrations converged at the reported values from Table III. In future works, ablation studies should be considered to fully consider the shuffling nuances with a sampled Pareto calibration approach.

IV. RESULTS

Arbitrarily constructed runs were conducted with output yielded from a previous experiment with a symbolic music LLM model. Candidate pairs were created in each run and share the precise prompting [16]. These experiment pairs aim to isolate the model by requiring specific conditions such as time signature, pitch signature and instrumentation. We

TABLE IV
RUN-STRATIFIED WITHIN-RUN LCS (ABSOLUTE-PITCH)

Stratum	n pairs	LCS	vs null
High	242	0.072 34	235×
Low	484	0.018 98	61×
H×H null	813 450	0.000 31	1×

hypothesize certain configurations are more opportune for invoking memorization conditions.

A. Instrumentation and Style Separation

Referencing Figure 1 and comparing the visual results to Table II, it is very difficult to assert that token-level memorization above the reported human baselines is occurring within the LLM generated output. The 13 unique pairs detected with Jaccard n -gram overlap confirm the complimentary nature of the metrics. The clear delineation between piano and string quartet samples indicates the importance of baselines for memorization research. The variation in style collapse and LCS scores contrasts greatly and highlights the importance of multiple metrics for the evaluation.

B. Within-Model Memorization

If LLMs are not directly memorizing token patterns, we hypothesize they may be gravitating towards a learned compositional style or high density training data region. One set of candidate outputs from a previous LLM run scored a mean LCS of 0.108; rating 350 times above the human-to-human baseline, illustrated in Table IV. Two sets from the generated output are identified and these distributions are statistically separable ($r = 0.205, p = 2.4 \times 10^{-13}$, 95% CI on mean difference [0.037, 0.069]).

With this data, memorization cannot be attributed to the self-similarity. Given the model’s capabilities to generate output with a best LCS score of 1.9%, we hypothesize that the training data may create more narrow conditions for unique generation. When comparing the interval-evaluations, three run groups maintain a high LCS that is transposition intolerant. Compared to six groups that fall below the threshold, two trends emerge with the compositional model style. In one instance, the model is incapable of producing key-invariant pieces and struggles to produce unique results. In another instance, the model is able to produce results that share the melodic structure of other songs but in a separate key. This data suggests that the model may have converted onto a specific key region rather than specific melodies within the training data.

C. Transposition Analysis

Interval-based tokenization [7] is employed to detect verbatim copying across the LLM’s output. Table V showcases the interval LCS means for each group alongside the absolute-pitch results. By analyzing the lifts, a better determination can be made regarding what aspects could be influencing the LLM’s generation patterns. Comparing TF-IDF cosine

TABLE V
ABSOLUTE PITCH VS INTERVAL LCS MEANS
COMPARED TO GROUP TF-IDF COSINE DISTANCES

Pair type	Abs	Interval	Lift	TF-IDF
LLM×LLM	0.009 47	0.015 87	+68%	0.702
LLM×FT	0.002 60	0.008 02	+208%	0.778
LLM×Train	0.000 51	0.001 24	+143%	0.843
H×H	0.000 31	0.000 31	0%	0.518

distances, LLM outputs are distinctly closer to the fine tuning data than the broader training data.

Inspecting Table V shows the absolute pitch and interval rating orders to be preserved across experiments. LLM similarity outweighs the target focus group of Fine-Tuned (FT) samples and exceeds that of the songs suspected of being within the pretraining data. When comparing the Mann-Whitney U significance results, LLM to FT is statistically separable from the null baseline under the interval experiment ($p < 0.001, r = 0.030$). When analyzing the high similarity runs to the human null baseline ($r = 0.303, p < 0.001$), the data generated suggests that the most articulable reason for human detection of memorization and subsequently plagiarism may be due to key-shifted reproduction data. In one notable case, two LLM samples in different time and key signatures were flagged exclusively by the interval metric; detecting transposition memorization.

D. Best Window LCS Significance Testing

Given the skew in song distribution for this test along with the various metrics employed, statistical testing is required to showcase the performances of each metric. Mann-Whitney U tests were conducted across all pairwise LCS distributions. TF-IDF distances confirm the fine tuning signal ($r = -0.339$ LLM to FT compared with LLM to Train). This test was selected over others to evaluate the distributions due to the results containing many zero results. All test are conducted with $\alpha = 0.05$ and effect sizes are reported as a correlation factor r . Confidence intervals (CI) of 95% are estimated using bootstrap resampling rate of 2,000. A brief Kruskal-Wallis H test ($H = 15, 275, p < 0.001$) confirms that the distributions are from distinct populations and allows us to compute the pairwise computations in Table VI. One consideration for future works will be to consider smaller sample sizes for significance testing.

Every comparison to the human null hypothesis within Table VI achieves $p < 0.001$, meaning the remaining pair types are distinguishable distributions. Given the large sample sizes, effect sizes must also be considered to mitigate the pitfalls of uneven distribution sampling with this technique of statistical significance testing. LLM to training and fine tuning data comparisons have detectable but statistically negligible memorization at the n -gram level. When comparing LLM output to itself, high comparison run-stratified samples yield a higher r value. The few samples that generate this signal are very strong in the large reference size.

V. CONCLUSION & FUTURE WORKS

Familiarity across LLM generated samples from this experiment can be attributed to the self-similarity finding more so than the training data memorization. It appears the base model, regardless of configuration, is prone to sharing similar elements across melodies. Shifts noted across intervals also indicate that the fine tuning data presented may have impacts to the similarity. Given that the model was previously fine tuned on this data, we hypothesize self-similarity may be induced because of this. While the model has not precisely committed copyright infringement, some could say that the model could be suspected of *self-plagiarism* in this particular instance. That said, there are multiple limitations that should be acknowledged that may impact the detection of memorization and should be revisited in future experiments.

The experiment composition is incomplete due to missing pretraining data sources; as evidenced by the negative Mann-Whitney U testing. PDMX and Maestro may be included in the pretraining and fine tuning data. The negative memorization results indicates no detectable memorization of the included data above the baseline, but it does not rule out memorization; merely the included samples were not memorized. Additionally, the n -gram Jaccard pre-filter implementation used for runtime cost savings may also exclude sparse n -gram overlaps. The filter itself was set fairly low, but there should be considerations to this. Lastly, the UMAP projection maintains a fixed-state for visual clarity. While these limitations can modify the specific levels of memorization detection, the trends observed and reported within this research suggest that this framework can be made capable of memorization detection.

Music generation research has accelerated rapidly in the last several years. Data quality, availability and capability of processing have enabled various researchers to extend far past what was previously possible. With that said, the technologies responsible for auditing and ensuring originality must also be developed at a similar pace to preserve human ingenuity alongside researcher efficacy. By incorporating this utility into generation platforms, researchers can have better guarantee to the originality of their systems. For musicians, they may be able to use this utility to evaluate the uniqueness of their work and protect their Intellectual Property further. With a lightweight utility like this, creating more unique contributions to both music and music generation research can be done with greater levels of confidence.

Future works should continue to study other models within this framework. Identifying pretraining data sources (or suspected sources) may be a beneficial line of research to test this system’s capabilities. By identifying highest-similarity pairs within the broader corpus, clear evidence can be aggregated to determine a music generation model’s capability to uniquely create. Additionally, studies can be conducted across generation models to determine overall uniqueness regardless of generation methodology. Symbolic music models can be evaluated on the same uniqueness criterion and referenced to the same training data; creating a level comparison between

TABLE VI
MANN-WHITNEY U TEST RESULTS (ABSOLUTE-PITCH LCS)

Comparison	Mean A	Mean B	p	r	Effect	95% CI (diff)
LLM×LLM vs H×H	0.009 47	0.000 31	$< 10^{-300}$	0.038	negl.	[0.0085, 0.0097]
LLM×FT vs H×H	0.002 60	0.000 31	5×10^{-172}	0.009	negl.	[0.0018, 0.0028]
LLM×Train vs H×H	0.000 51	0.000 31	2×10^{-11}	0.001	negl.	[0.000004, 0.00028]
LLM×FT vs LLM×Train	0.002 60	0.000 51	10^{-47}	0.008	negl.	—
High runs vs Low runs	0.072 34	0.018 98	2×10^{-13}	0.205	small	[0.037, 0.069]
High runs vs H×H	0.072 34	0.000 31	$< 10^{-300}$	0.280	small	—

otherwise closed systems. Finally, future works should also consider testing with specific samples that are clear examples of memorization or plagiarism.

REFERENCES

- [1] Peter Knees and Markus Schedl, “A survey of music similarity and recommendation from music context data,” *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, vol. 10, no. 1, pp. 1–21, 2013.
- [2] Mohamadreza Sheikh Fathollahi and Farbod Razzazi, “Music similarity measurement and recommendation system using convolutional neural networks,” *International Journal of Multimedia Information Retrieval*, vol. 10, no. 1, pp. 43–53, 2021.
- [3] Brian McFee, Luke Barrington, and Gert Lanckriet, “Learning content similarity for music recommendation,” *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 20, no. 8, pp. 2207–2218, 2012.
- [4] Jacob Edward Galajda and Kien Hua, “Automated thematic composer classification using segment retrieval,” in *2024 IEEE 7th International Conference on Multimedia Information Processing and Retrieval (MIPR)*. IEEE, 2024, pp. 162–168.
- [5] Jacob Edward Galajda and Kien Hua, “Neural-base music generation for intelligence duplication,” *arXiv preprint arXiv:2310.13691*, 2023.
- [6] Asli Saner and Min Chen, “Music similarity detection through comparative imagery data,” *Applied Sciences*, vol. 15, no. 14, pp. 7706, 2025.
- [7] Roberto De Prisco, Delfina Malandrino, Gianluca Zaccagnino, and Rocco Zaccagnino, “Fuzzy vectorial-based similarity detection of music plagiarism,” in *2017 IEEE international conference on fuzzy systems (FUZZ-IEEE)*. IEEE, 2017, pp. 1–6.
- [8] Yumin Kim and Seonghyeon Go, “Segment transformer: Ai-generated music detection via music structural analysis,” in *2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*. IEEE, 2025, pp. 664–669.
- [9] Sansiri Tarnpradab, Fereshteh Jafariakinabad, and Kien A. Hua, “Improving online forums summarization via hierarchical unified deep neural network,” 2021.
- [10] Fereshteh Jafariakinabad, “Machine learning techniques for topic detection and authorship attribution in textual data,” (2021). *Electronic Theses and Dissertations*, vol. 884., 2020.
- [11] Delfina Malandrino, Roberto De Prisco, Mario Ianulardo, and Rocco Zaccagnino, “An adaptive meta-heuristic for music plagiarism detection based on text similarity and clustering,” *Data Mining & Knowledge Discovery*, vol. 36, no. 4, 2022.
- [12] Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel, “Extracting training data from large language models,” in *30th USENIX Security Symposium (USENIX Security 21)*. Aug. 2021, pp. 2633–2650, USENIX Association.
- [13] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang, “Quantifying memorization across neural language models,” in *The Eleventh International Conference on Learning Representations*, 2023.
- [14] R. Singh and S. Singh, “Text similarity measures in news articles by vector space model using NLP,” *J. Inst. Eng. India Ser. B*, vol. 102, 2021.
- [15] Abdul Wahab Qurashi, Violeta Holmes, and Anju P. Johnson, “Document processing: Methods for semantic text similarity analysis,” in *2020 International Conference on INnovations in Intelligent SysTems and Applications (INISTA)*, 2020, pp. 1–6.
- [16] Jacob Edward Galajda, Kien Hua, et al., “Deep composer: Improving the string quartet music generation task,” in *2025 IEEE 8th International Conference on Multimedia Information Processing and Retrieval (MIPR)*. IEEE, 2025, pp. 86–92.
- [17] Weihai Xu, Julian McAuley, Taylor Berg-Kirkpatrick, Shlomo Dubnov, and Hao-Wen Dong, “Generating symbolic music from natural language prompts using an llm-enhanced dataset,” *arXiv preprint arXiv:2410.02084*, 2024.
- [18] Shanga Wu, Dingyao Yu, Xu Tan, and Maosong Sun, “Clamp: Contrastive language-music pre-training for cross-modal symbolic music information retrieval,” 2023.
- [19] Qiuqiang Kong, Keunwoo Choi, and Yuxuan Wang, “Large-scale midi-based composer classification,” *ArXiv*, vol. abs/2010.14805, 2020.
- [20] Matthew Cooper and Jonathan Foote, “Summarizing popular music via structural similarity analysis,” in *2003 IEEE workshop on applications of signal processing to audio and acoustics (IEEE Cat. No. 03TH8684)*. IEEE, 2003, pp. 127–130.
- [21] Tao Li and Mitsunori Ogihara, “Content-based music similarity search and emotion detection,” in *2004 IEEE International Conference on Acoustics, Speech, and Signal Processing*. IEEE, 2004, vol. 5, pp. V–705.
- [22] Alexander Lerch, *Music Similarity Detection and Music Genre Classification*, pp. 317–335, Wiley-IEEE Press, 2023.
- [23] Juan P Bello, “Measuring structural similarity in music,” *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 19, no. 7, pp. 2013–2025, 2011.
- [24] Seonghyeon Go, “Real-world music plagiarism detection with music segment transcription system,” in *2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2025, pp. 276–281.
- [25] Tongyu Lu, Charlotta-Marlena Geist, Jan Melechovsky, Abhinaba Roy, and Dorien Herremans, “Melodysim: measuring melody-aware music similarity for plagiarism detection,” *arXiv preprint arXiv:2505.20979*, 2025.
- [26] Carlos Hernandez-Olivan and Jose R. Beltran, “Symbolic music structure analysis with graph representations and changepoint detection methods,” *Neurocomputing*, vol. 666, pp. 132208, 2026.
- [27] Curtis Hawthorne, Andriy Stasyuk, Adam Roberts, Ian Simon, Cheng-Zhi Anna Huang, Sander Dieleman, Erich Elsen, Jesse Engel, and Douglas Eck, “Enabling factorized piano music modeling and generation with the MAESTRO dataset,” in *International Conference on Learning Representations*, 2019.
- [28] Mike Cuthbert and Christopher Ariza, “Music21: A toolkit for computer-aided musicology and symbolic music data,” in *International Society for Music Information Retrieval Conference*, 2010.
- [29] Hwei-Jen Lin, Hung-Hsuan Wu, and Chun-Wei Wang, “Music matching based on rough longest common subsequence,” *J. Inf. Sci. Eng.*, vol. 27, no. 1, pp. 95–110, 2011.
- [30] Luciano da F. Costa, “Further generalizations of the jaccard index,” 2021.
- [31] Leland McInnes, John Healy, and James Melville, “Umap: Uniform manifold approximation and projection for dimension reduction,” 2020.